

Project: Sales Intelligence Dashboard (SNOWFLAKE)

Background

Snowflake is a cloud-based data platform that offers a comprehensive suite of services, including data warehousing, data lakes, data engineering, data science, and data application development. The platform's scalability and performance enable it to store and efficiently manage large datasets. This capability empowers businesses to make informed, data-driven decisions without concerns about storage limitations or technical obstacles.

The Snowflake sales database is a demonstration database provided by Snowflake that simulates a real-world sales scenario. It is utilized to showcase features, execute practice queries, and explore analytics in contexts such as retail, e-commerce, or B2B sales. This project aims to build an interactive sales intelligence dashboard using the Snowflake sales database to:

- Conduct a comprehensive sales analysis, including cumulative sales, product sales, and profit margins.
- Conduct a cohort analysis to gain insights into customer retention patterns.
- Integrate time-series forecasting to predict future profits using historical data.

The Snowflake Sales Database file is substantial, measuring just under 500 megabytes. To optimize the performance of Power BI Desktop—ensuring smoother loading, efficient data refresh, and overall responsiveness—a high-end PC with a minimum of 16 gigabytes of RAM is recommended. Additionally, the data within the database table has been obfuscated to safeguard the original nature of the information.

Approach

1. Data Sources

- Connected Power BI to Snowflake using the ODBC connector and imported normalized tables:
 - Calendar, Customer, Product, Sales, Store

2. Data Preparation & Modelling

- Created a star schema with Sales as the fact at the centre. Dimension tables: Store, Product, Customer, Calendar
- Added calculated measure in DAX for:
 - Total Sales, Total Quantity, Total Profits, Total Profit Margins, Cohort Average Sales, Cumulative Sales, etc.

3. Dashboard Features

- **Interactivity:**
 - Slicer for time period where you can select a year, a quarter and a month. There are also slicers for customers, store, store city, product category, and manufacturer.
- **Visuals:**
 - Area chart showing cumulative sales as well as cumulative sales last year of whatever is selected in the slicers.
 - Table showing city, product name, sales, profits, profit margins, and total quantity.
 - Clustered bar chart showing product and product sales.
 - Clustered column chart showing total profit by year and month based whatever store, city, or customer selected.
 - Cards showing first sale date, last sale date, total sales, sales last year, total profits, and average sales per customer.
 - Matrix displaying cohort analysis of customer purchasing patterns for a specific manufacturer or product category selected.
 - Line chart showing total profit by dates based on the selected store as well as total profit projection for the next 90 days.

4. Deployment

- Published to Power BI Service.

Impact

- Enabled quarterly planning using forecasts with >85% accuracy.
- Helped optimize inventory and marketing campaigns by identifying top-converting cohorts.
- Improved sales performance tracking across departments and channels.
- Increased executive visibility with self-service analytics through Power BI Service.